

Project 1: Building an Agentic Corrective RAG System in LangGraph

Instructor

Dipanjan Sarkar

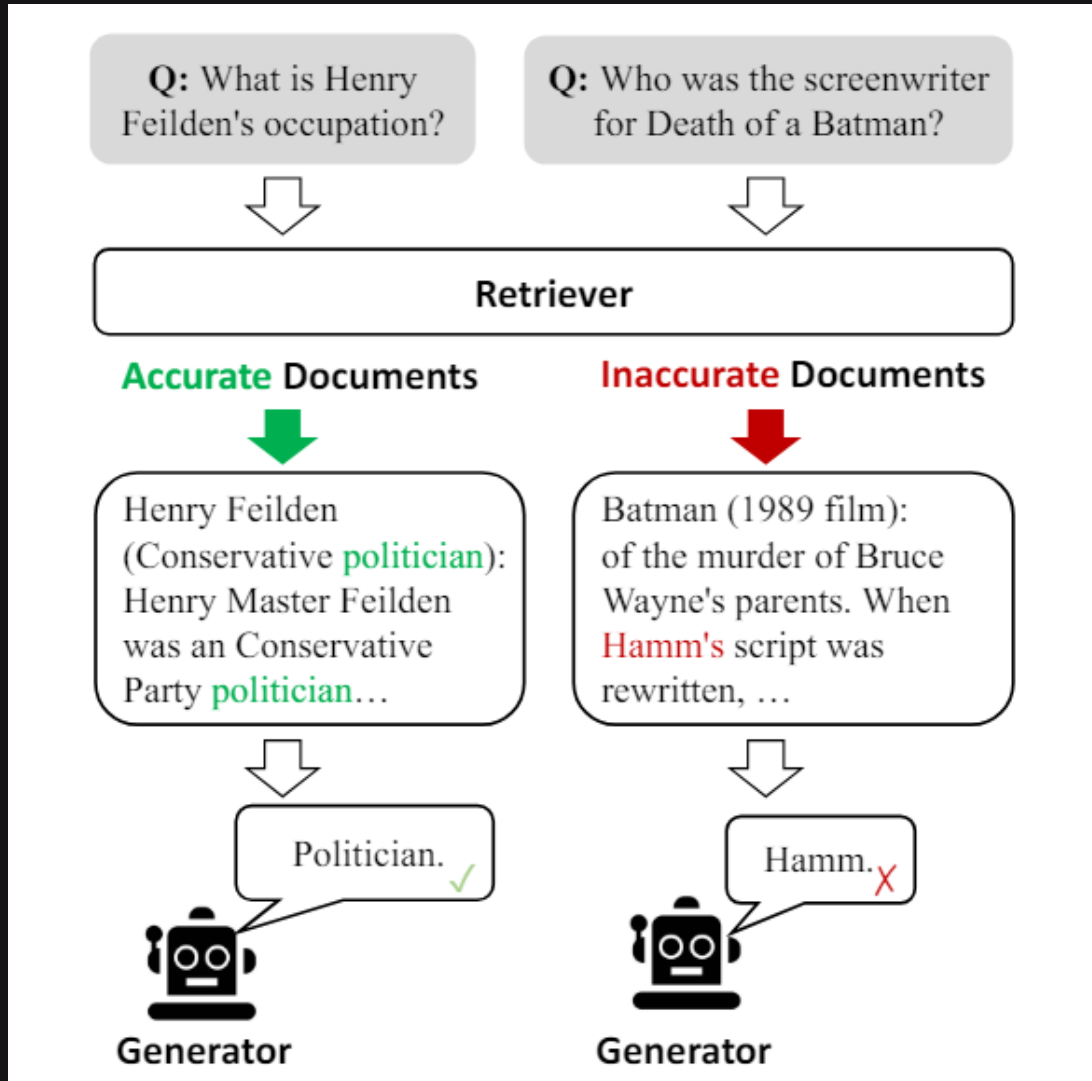
Head of Community & Principal AI Scientist at Analytics Vidhya

Google Developer Expert - ML & Cloud Champion Innovator

Published Author

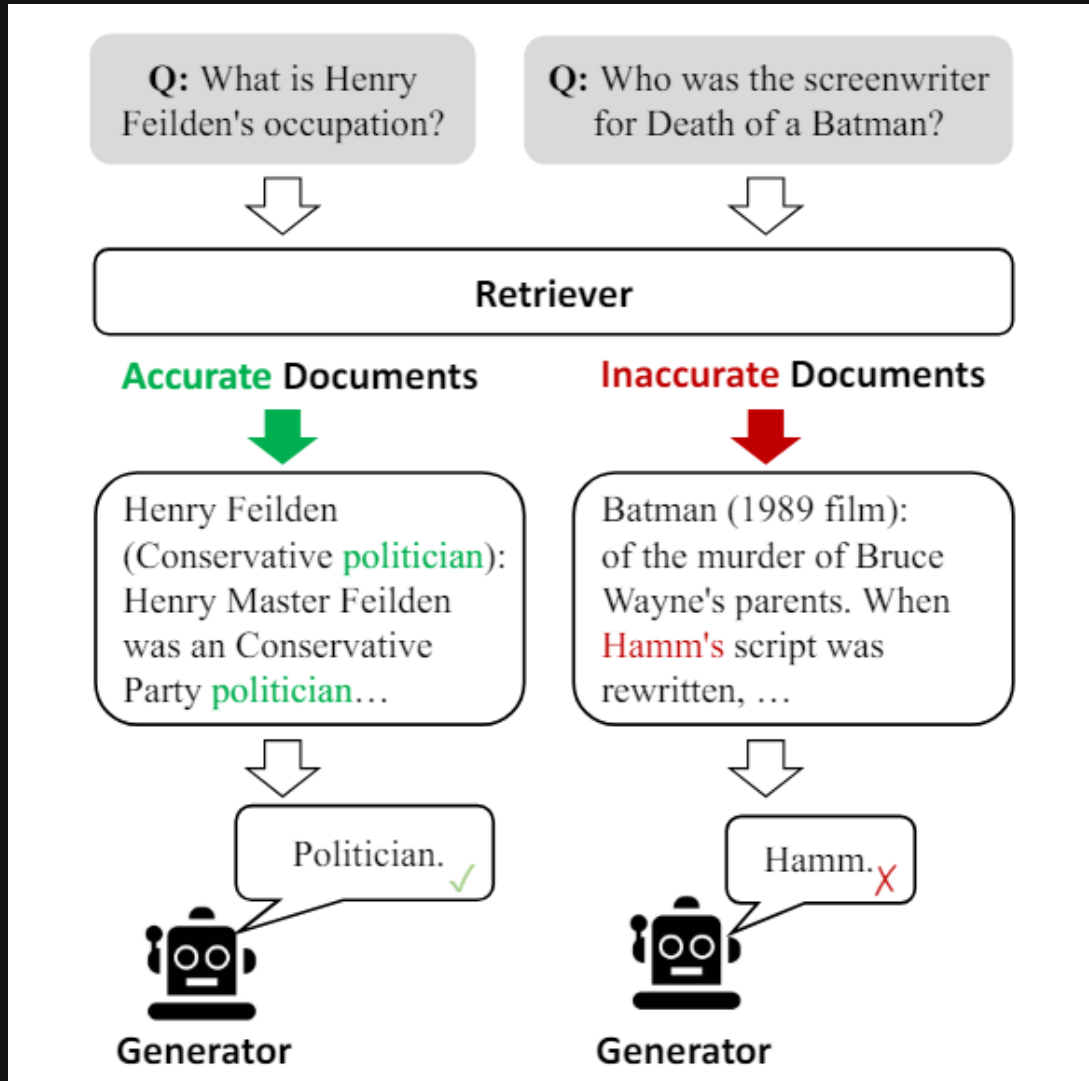


Key challenges in RAG systems?



- **Outdated or Low-Quality Retrieval:**
 - Compromises the performance and accuracy of RAG systems.
- **Misinformation Risk:**
 - Leads to the generation of incorrect or misleading information by LLMs.

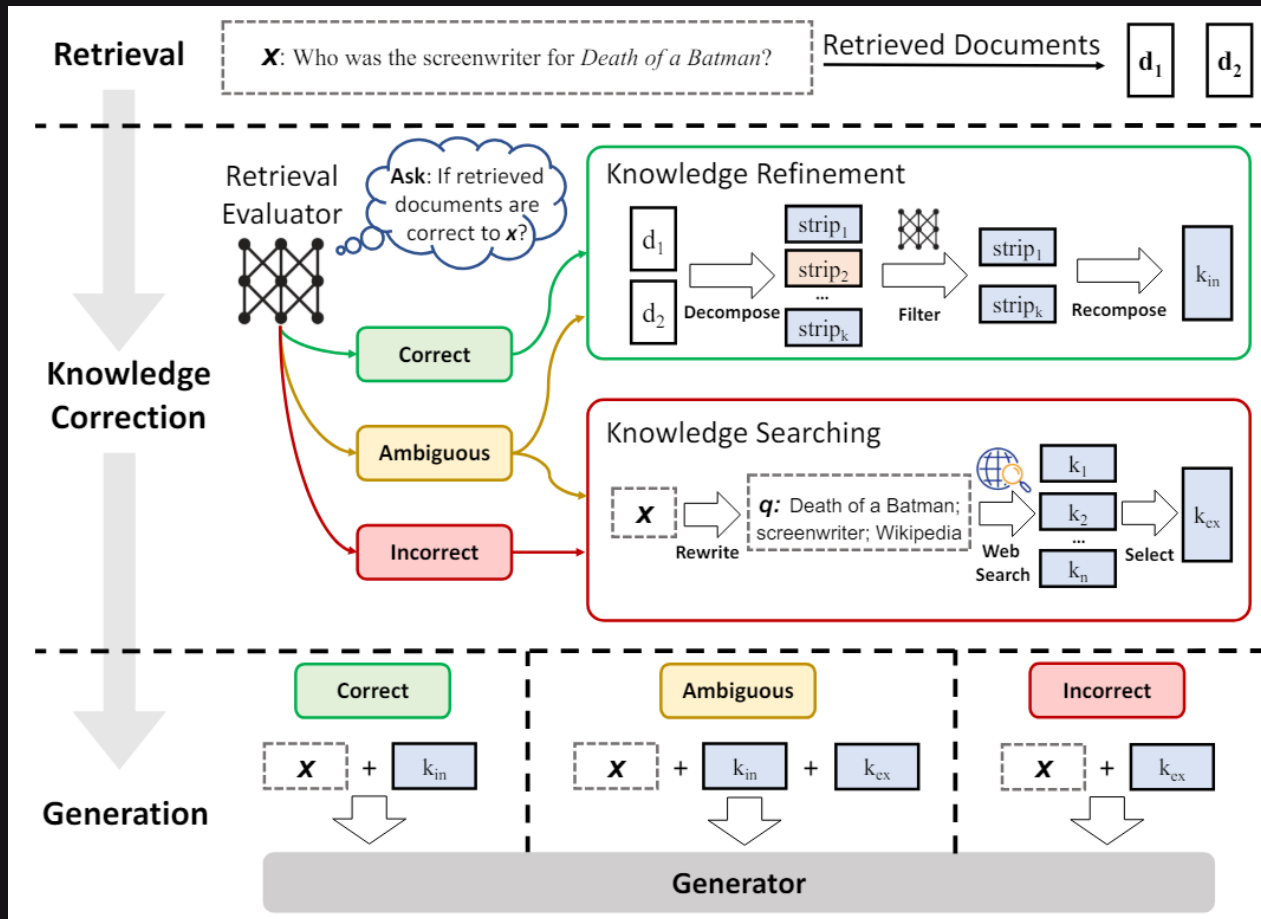
Key challenges in RAG systems?



- **Proposed Solution - CRAG:**

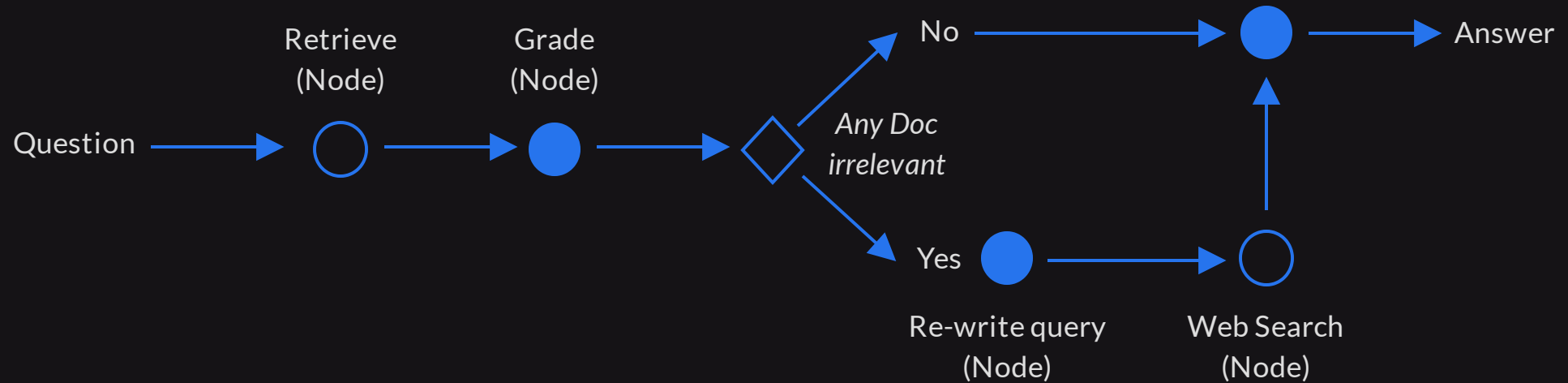
- "Corrective Retrieval Augmented Generation" aims to mitigate these issues by improving retrieval accuracy.

Corrective RAG Workflow proposed in the CRAG paper



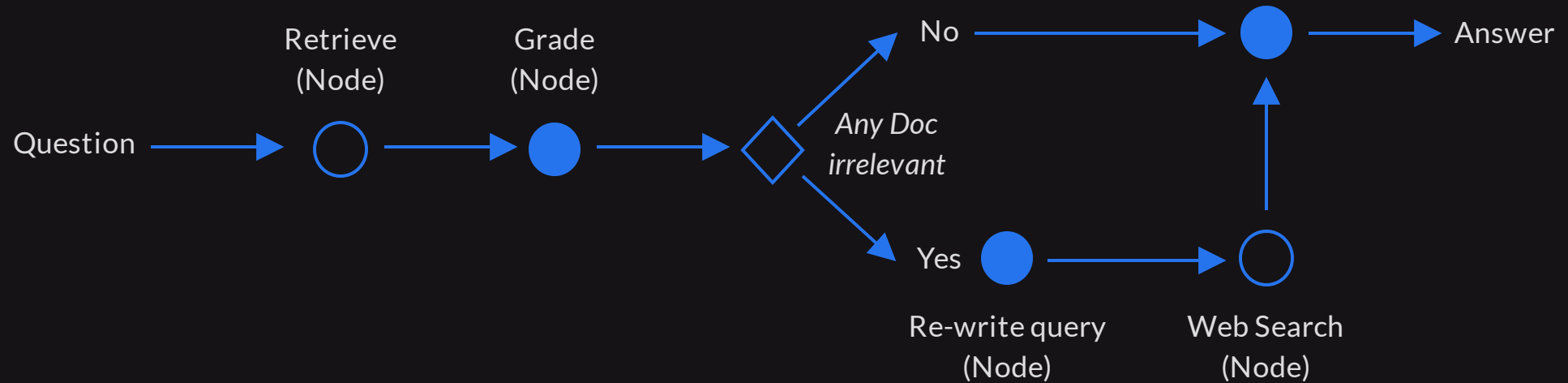
- Step 1:
 - Retrieve context documents from vector database from the input query
- Step 2:
 - Use an LLM to check if retrieved documents are relevant to input question
- Step 3:
 - If all documents are relevant (Correct), no specific action needed
- Step 4:
 - If some or all documents are not relevant (Ambiguous OR Incorrect), rephrase the query and search the web to get relevant context information
- Step 5:
 - Send rephrased query and context documents or information to the LLM for response generation

Agentic Corrective RAG Workflow to implement



- **Step 1 - Retrieve Node**
 - Retrieves context documents from vector database from input query
- **Step 2 - Grade Node**
 - Use an LLM to grade if retrieved documents are relevant to input question - yes or no
- **Step 3A - Generate Answer Node**
 - If all documents are relevant (all 'yes'), send it to an LLM for response generation

Agentic Corrective RAG Workflow to implement



- **Step 3B - Rewrite Query Node**
 - If some or all documents are not relevant (at least one 'no'), rephrase the query
- **Step 3C - Web Search Node**
 - Search the web to get context information using rephrased query
- **Step 3D - Generate Answer Node**
 - Send rephrased query and context documents or information to the LLM for response generation

Detailed Agentic Corrective RAG Workflow

